Econ 5551

Multiple Hypothesis Testing

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Multiple Hypothesis Testing

- Suppose that you wished to evaluate the performance of each of J different fund managers. Suppose for each fund manager $j \in \{1, \ldots, J\}$,
  - We observe $S_j \sim \text{Binomial}(\theta_j, 20)$, where θ_j is the probability in a given year that the manager's fund outperforms the benchmark.
  - Will test at the 0.05-level the null that $H_{0,j} : \theta_j = 0.5$ vs $H_{1,j} : \theta_j > 0.5$.

  Our goal is to determine which (if any) of the fund managers can systematically outperform the benchmark.

Multiple Hypothesis Testing

- Suppose that you wished to evaluate the performance of each of J different fund managers. Suppose for each fund manager $j \in \{1, \ldots, J\}$,
  - We observe $S_j \sim \text{Binomial}(\theta_j, 20)$, where θ_j is the probability in a given year that the manager's fund outperforms the benchmark.
  - Will test at the 0.05-level the null that $H_{0,j} : \theta_j = 0.5$ vs $H_{1,j} : \theta_j > 0.5$.

  Our goal is to determine which (if any) of the fund managers can systematically outperform the benchmark.

- We are thus testing a *family* of J null hypotheses, $H_{0,1}$,...,$H_{0,J}$. Testing a family of $J \geq 2$ null hypothesis is called a *multiple hypothesis testing problem*.
  - How is this problem different from a joint test of the single null that $H_0 : \theta_j = 0.5$ for all j, vs $H_{1,j} : \theta_j > 0.5$ for some j?

FWER

Definition (FWER)

The *Family Wise Error Rate (FWER)* is the probability of one or more false rejections among the family of hypotheses.

- In our example, the FWER is probability of concluding that at least one manager is able to systematically outperform the market who is not actually able to do so.
- In general, FWER depends on . . .
  - how many null hypotheses are true
    - For example, FWER=0 if no null is true.
  - the dependence across the test statistics.
    - In our fund example, dependence across the S_j.

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.

(*by definition of FWER*)

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$$\begin{aligned}\text{FWER} &= \Pr[\text{Reject } H_{0,j} \text{ for some j}] \\ &= 1 - \Pr[\text{Fail to Reject } H_{0,j} \text{ for any j}]\end{aligned}$$

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(*by complement rule*)

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(*by independence of test statistics*)

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&= 1 - \Pr[\text{Fail to Reject } H_{0,j} \text{ for any j}] \\
&= 1 - \Pr[\text{Fail to Reject } H_{0,1}] \times \ldots \times \Pr[\text{Fail to Reject } H_{0,J}] \\
&= 1 - (1-\alpha)^J.
\end{aligned}$$

(*by each test being of level* α)

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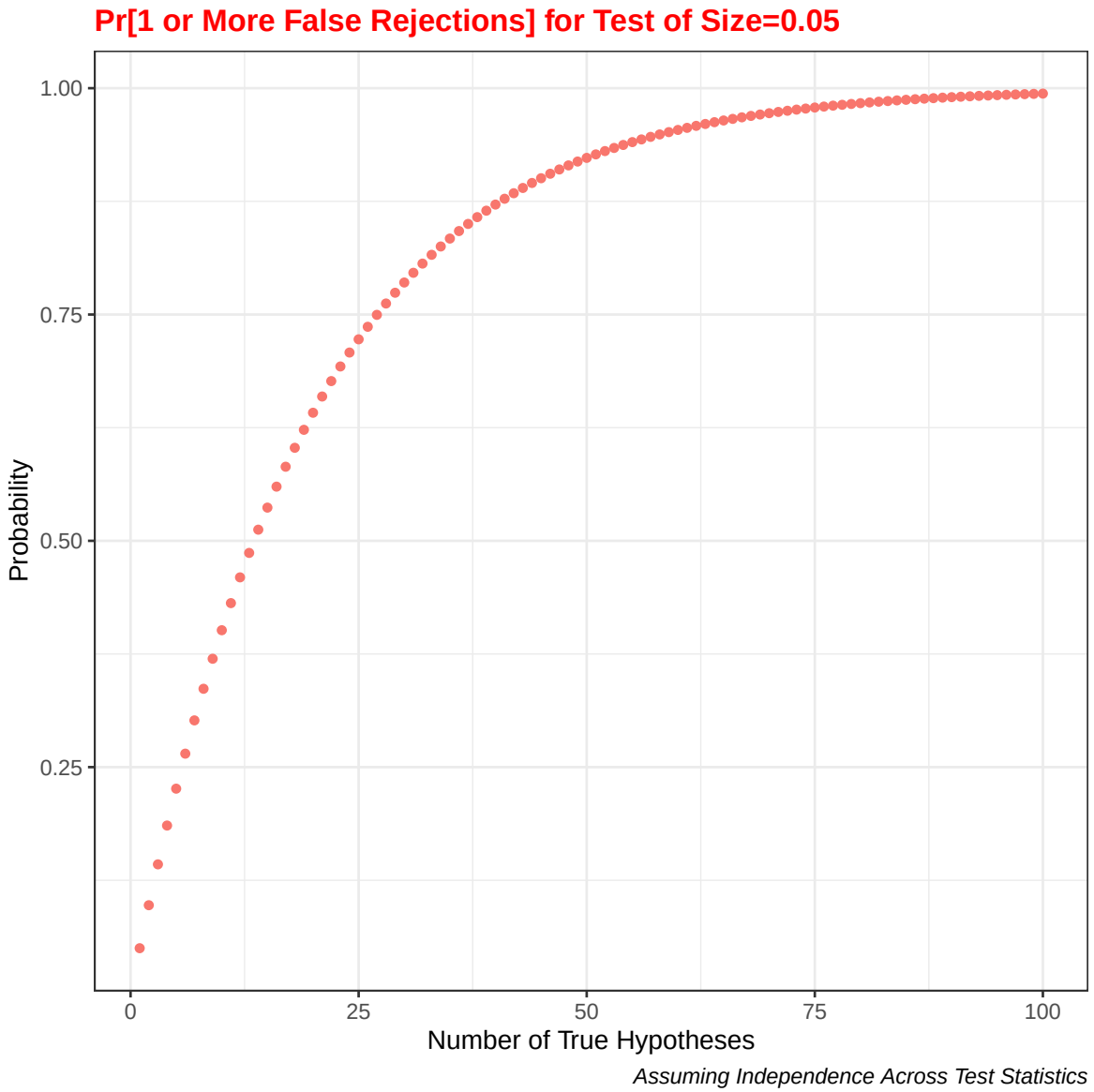
- For example, suppose that we separately perform J hypothesis tests, each at level α, that the test statistics are mutually independent, and that all J null hypotheses are true. Then

$$\begin{aligned}
\text{FWER} &= \Pr\big[\text{Reject } H_{0,j} \text{ for some j}\big] \\
&= 1 - \Pr\big[\text{Fail to Reject } H_{0,j} \text{ for any j}\big] \\
&= 1 - \Pr\big[\text{Fail to Reject } H_{0,1}\big] \times \ldots \times \Pr\big[\text{Fail to Reject } H_{0,J}\big] \\
&= 1 - \big(1 - \alpha\big)^J.
\end{aligned}$$

For example, for an 0.05 level test, and $J = 100$, $1 - \big(0.95\big)^{100} = 0.994$.



Probability of at Least One False Rejection for 0.05 Level Test

No. True Nulls	Prob.
1	0.050
5	0.226
10	0.401
50	0.923
100	0.994

Assuming independent test statistics.

FWER

Definition (Strong Control of FWER)

A testing procedure *strongly controls* the FWER at level α if FWER $\leq \alpha$ under every possible constellation of true and false null hypotheses.

- Strong control of FWER contrasts with *weak control*, which only requires FWER $\leq \alpha$ when all J nulls are simultaneously true.
- Instead of designing our testing procedure so that the probability of falsely rejecting each null is at most α, we can design our testing procedure for strong control of the FWER.
- Let p_j denote the p-value for testing $H_{0,j}$. Multiple hypothesis testing procedures typically use as inputs $(p_1, \ldots, p_J)$, typically can be interpreted as "correcting" the p-values for the multiplicity of hypotheses. The simplest procedure is the *Bonferroni Procedure* . . .

Bonferroni

Definition (Bonferroni Procedure)

Reject those $H_{0,j}$ for which $p_j \leq \alpha/J$.

Theorem

The Bonferroni procedure strongly controls the FWER at level α.

Proof:

- Fix a data-generating process P, let I denote the set of true nulls under P, and let $|I| = K \leq J$.
- Let V denote the number of false rejections.
- For each $i \in I$, let
$$A_i = \{p_i \leq \alpha/J\} = \{H_{0,i} \text{ is rejected}\}.$$
- Then
$$\Pr(V \geq 1) = \Pr\left(\bigcup_{i \in I} A_i\right) \leq \sum_{i \in I} \Pr(A_i) \leq \sum_{i \in I} \frac{\alpha}{J} = K\frac{\alpha}{J} \leq \alpha.$$

Bonferroni (cont'd)

Definition (Bonferroni Procedure)

Reject those $H_{0,j}$ for which $p_j \leq \alpha/J$.

Theorem

The Bonferroni procedure strongly controls the FWER at level α.

- More generally, for any subset S with $|S| = m$, the same Bonferroni inequality also yields a level-α test of the intersection null

$$H_S = \bigcap_{j \in S} H_{0,j},$$

which rejects when

$$\min_{j \in S} p_j \leq \alpha/|S|.$$

This is a joint test of H_S. This feature of the Bonferroni inequality will be useful later for understanding the Holm's and Romano-Wolf testing procedures.

Bonferroni (cont'd)

Definition (Bonferroni Procedure)

Reject those $H_{0,j}$ for which $p_j \leq \alpha/J$.

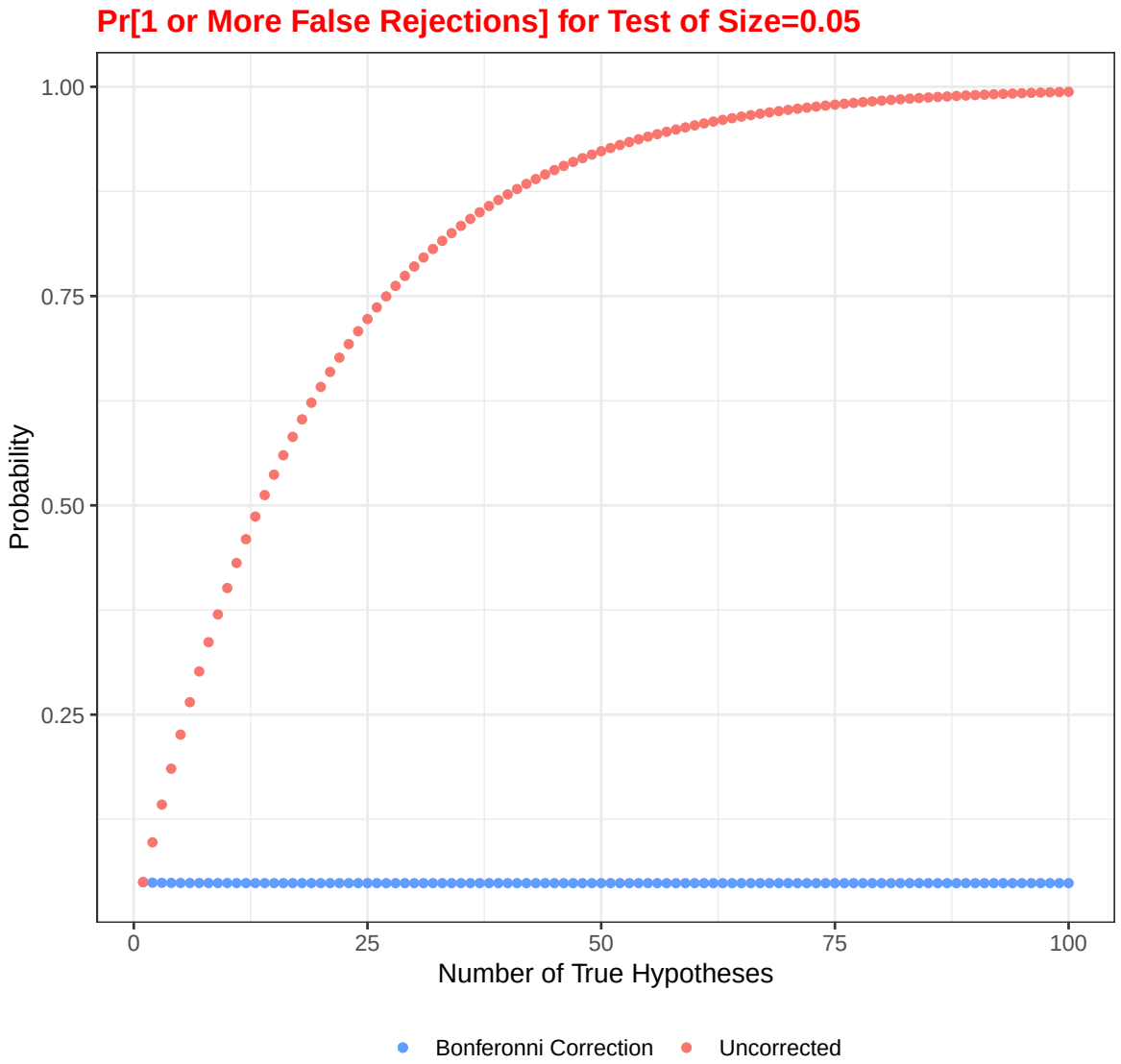
Theorem

The Bonferroni procedure strongly controls the FWER at level α.

Recall that to test the null $H_{0,j}$ in isolation, we would reject if $p_j \leq \alpha$.

- Can view Bonferroni procedure as correcting the p-value to be $J \times p_j$ and rejecting when corrected p-value is less than α.
- Can implement in **R** with `p.adjust` using `method=bonferroni`.
- Much harder to reject with the correction, especially for J large. For example, if $\alpha = 0.05$ and $J = 100$, then would reject without the correction when $p_j < 0.05$ and with the correction when $p_j < 0.0005$.

Bonferroni (cont'd)



Assuming Independence Across Test Statistics

Probability of at Least One False Rejection for 0.05 Level Test

No. True Nulls	Uncorr.	Bonf. Corr.
1	0.050	0.050
5	0.226	0.049
10	0.401	0.049
50	0.923	0.049
100	0.994	0.049

Assuming independent test statistics.

Bonferroni (cont'd)

Downsides of correction:

- Much less power, higher probability of type II error for false hypotheses.
- Sensitivity to how many hypotheses are included in the family.
- Sensitive to manipulation:
  - If wish to reject null, include fewer nulls in family, admit to search over smaller set of null hypotheses.
  - If wish to not reject null, include more nulls in family.

Bonferroni (cont'd)

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  - If wish to not reject null, include more nulls in family.

For example, consider fund example, with $J \geq 2$, and

- $\theta_1 = \theta_2 = 0.7$, so that $H_{0,1}$ and $H_{0,2}$ are false, and
- $\theta_3 = \theta_4 = \ldots \theta_J = 0.5$, so that the remaining $J - 2$ nulls are true;
- S_j mutually independent across j.

Bonferroni (cont'd)

Power with and Without Bonferroni Correction

$H_{0,1}$ and $H_{0,2}$ False, Remaining Nulls True

	$\Pr[\text{Rej. } H_{0,1} \cup \text{Rej. } H_{0,2}]$		$\Pr[\text{Rej. } H_{0,1} \cap \text{Rej. } H_{0,2}]$	
No. Nulls	Uncorrected	Corrected	Uncorrected	Corrected
2	0.66	0.66	0.17	0.17
3	0.66	0.42	0.17	0.06
10	0.66	0.20	0.17	0.01
50	0.66	0.07	0.17	0

Tests at 0.05 level, assuming independence across test statistics.

Alternative Procedure: Holm Step-Down Method

Definition (Holm's Procedure at Level α)

1. Let $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(J)}$ denote the ordered p-values, with corresponding hypotheses $H_{0,(1)}, \ldots, H_{0,(J)}$.
2. Start with $k = 1$.
3. Compare $p_{(k)}$ to $\alpha/(J+1-k)$. If $p_{(k)} \leq \alpha/(J+1-k)$, reject $H_{0,(k)}$ and continue to $k+1$.
4. If $p_{(k)} > \alpha/(J+1-k)$, stop and do not reject $H_{0,(k)}, \ldots, H_{0,(J)}$.

Theorem

Holm's procedure strongly controls the FWER at level α under arbitrary dependence and rejects every hypothesis that the Bonferroni procedure rejects (and possibly more).

Proof that Holm's procedure controls FWER

- Proof strategy: Holm is a stepdown version of Bonferroni intersection tests. We identify the first true null in the ordered list, then show that any false rejection implies rejection of the Bonferroni level-α test of the intersection of the true nulls.
- Let $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(J)}$ denote the ordered realized p-values, with corresponding hypotheses $H_{0,(1)}, \ldots, H_{0,(J)}$.
- Fix a data-generating process P, let I denote the set of true nulls under P, and let $|I| = K \leq J$.
- Let k_0 be the position of the first true null in the ordered list:

$$k_0 = \min\{k : H_{0,(k)} \text{ is true}\},$$

  so that

$$p_{(k_0)} = \min_{i \in I} p_i.$$

  If Holm makes any false rejection, it must reject $H_{0,(k_0)}$.
- At most $J - K$ false nulls can appear before the first true null, so

$$k_0 \leq J - K + 1 \qquad \Rightarrow \qquad K \leq J - k_0 + 1.$$

Proof that Holm's procedure controls FWER (cont'd)

- If Holm rejects $H_{0,(k_0)}$, then by the step-down rule and that $K \leq J - k_0 + 1$,

$$p_{(k_0)} \leq \frac{\alpha}{J - k_0 + 1} \leq \frac{\alpha}{K}.$$

- Therefore,

$$\{\text{at least one false rejection}\} \subseteq \left\{ \min_{i \in I} p_i \leq \frac{\alpha}{K} \right\}.$$

- The event

$$\left\{ \min_{i \in I} p_i \leq \frac{\alpha}{K} \right\}$$

is exactly the rejection event of the Bonferroni level-α test of the intersection null

$$\bigcap_{i \in I} H_{0,i}.$$

Proof that Holm's procedure controls FWER (cont'd)

- Since each true-null p-value is valid,

$$\Pr\left(p_i \leq \frac{\alpha}{K}\right) \leq \frac{\alpha}{K} \qquad \text{for each } i \in I.$$

- Therefore, with V denoting the number of false rejections,

$$\begin{aligned}
\Pr(V \geq 1) &\leq \Pr\left(\min_{i \in I} p_i \leq \frac{\alpha}{K}\right) \\
&= \Pr\left(\bigcup_{i \in I} \left\{p_i \leq \frac{\alpha}{K}\right\}\right) \\
&\leq \sum_{i \in I} \Pr\left(p_i \leq \frac{\alpha}{K}\right) \\
&\leq \sum_{i \in I} \frac{\alpha}{K} \\
&= \alpha.
\end{aligned}$$

- So Holm's procedure strongly controls the FWER at level α.

Alternative Procedure: Holm Step-Down Method (cont'd)

Holm Step-Down Method:

- Holm's step-down method also strongly controls the FWER and rejects at least as many hypotheses as the Bonferroni correction.
- Can implement in **R** with `p.adjust` using `method=holm`.
- Typically, slightly more power than Bonferroni method.
- Still sensitive to how many hypotheses are included in the family and to manipulation.

Alternative Procedure: Holm Step-Down Method (cont'd)

Holm Step-Down Method:

- Holm's step-down method also strongly controls the FWER and rejects at least as many hypotheses as the Bonferroni correction.
- Can implement in **R** with `p.adjust` using `method=holm`.
- Typically, slightly more power than Bonferroni method.
- Still sensitive to how many hypotheses are included in the family and to manipulation.

If the number of hypotheses J is very large, FWER control may be too stringent, and procedures that control FWER can have low power. In such settings, one common alternative is to control the false discovery rate (FDR) (Benjamini and Hochberg, 1995), defined as expected proportion of rejections that are false (defined as zero when nothing is rejected). FDR equals FWER when all null hypotheses are true, but is less stringent otherwise.

Example: PROGRESA

	p-values	
Grade	boys	girls
1	0.347	0.609
2	0.005	0.931
3	0.714	0.339
4	0.0004	0.025
5	0.297	0.076
6	0.002	0.007
7	0.012	0.0001
8	0.558	0.628
9	0.276	0.681
10	0.267	0.657

Randomization inference conditional on sex and grade of child:

- Using $3,000$ simulations from the $\binom{491}{308}$ possible treatment assignments.
- Showing simulated p-values for the sharp null of no treatment effect.
- In the slides below, boys and girls are treated as separate families of 10 grade-specific hypotheses each. If instead we defined one family of all 20 sex-grade cells, the adjustment would differ – recall our earlier point about sensitivity to family definition.

Example: PROGRESA

Table 0.1: P-Values: Boys, Testing Sharp Null of No Treatment Effect

	No Correction	Bonferroni	Holm's
1	0.347	1	1
2	0.005	0.054	0.043
3	0.714	1	1
4	0.0004	0.004	0.004
5	0.297	1	1
6	0.002	0.016	0.014
7	0.012	0.121	0.085
8	0.558	1	1
9	0.276	1	1
10	0.267	1	1

Example: PROGRESA

Table 0.2: P-Values: Girls, Testing Sharp Null of No Treatment Effect

	No Correction	Bonferroni	Holm's
1	0.609	1	1
2	0.931	1	1
3	0.339	1	1
4	0.025	0.249	0.199
5	0.076	0.762	0.533
6	0.007	0.068	0.061
7	0.0001	0.001	0.001
8	0.628	1	1
9	0.681	1	1
10	0.657	1	1

Limitation of Holm: Ignores Dependence Structure

- Holm's procedure strongly controls FWER $\leq \alpha$ for *any* dependence structure among the test statistics.
- That robustness can be conservative because Holm does not use the joint dependence structure among the test statistics.
- In many applications—e.g., multiple outcomes measured on the same individuals—the test statistics are substantially correlated.
- Consequence: Holm's adjusted p-values can be larger than necessary, so the procedure may reject fewer hypotheses than a valid dependence-aware method.
- Key insight: If we can estimate the *joint* null distribution of the test statistics, we may be able to gain power while still controlling FWER.

Romano–Wolf Step-Down Procedure

- Romano and Wolf (2005, *JASA*; 2005b, *Econometrica*) propose resampling-based step-down procedures that use the joint dependence structure of the test statistics rather than Bonferroni-type worst-case bounds.
- Key idea: At each step of the algorithm, use resampling to approximate the null distribution of the maximum test statistic over the hypotheses that have not yet been rejected.
- This is a natural extension of the bootstrap/permutation logic we have already studied:
  - For a single hypothesis, compare $|T_n|$ to the resampled distribution of $|T_n^*|$.
  - For J hypotheses, compare the current test statistic to the resampled distribution of the maximum over the currently unrejected hypotheses.
  - This automatically incorporates the correlation across tests.

Romano–Wolf: Algorithm

Definition (Romano–Wolf Step-Down Procedure at Level α)

1. Compute test statistics $t_1, \ldots, t_J$ such that larger values indicate stronger evidence against the null, and order them as $t_{(1)} \geq t_{(2)} \geq \cdots \geq t_{(J)}$.
2. For $b = 1, \ldots, B$, generate one null-imposed resample and compute the full vector $t_1^{*(b)}, \ldots, t_J^{*(b)}$ on that same draw.
3. Start with $k = 1$.
4. Let $S_k = \{(k), \ldots, (J)\}$ denote the hypotheses not yet rejected, and compute the stagewise p-value
$$\tilde{p}_{(k)} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\left\{ \max_{j \in S_k} t_j^{*(b)} \geq t_{(k)} \right\}.$$
5. Define adjusted p-values recursively by
$$\hat{p}_{(1)}^{RW} = \tilde{p}_{(1)}, \qquad \hat{p}_{(k)}^{RW} = \max\{\hat{p}_{(k-1)}^{RW}, \tilde{p}_{(k)}\}, \quad k = 2, \ldots, J.$$
6. If $\hat{p}_{(k)}^{RW} \leq \alpha$, reject $H_{0,(k)}$ and continue to $k + 1$.
7. If $\hat{p}_{(k)}^{RW} > \alpha$, stop and do not reject $H_{0,(k)}, \ldots, H_{0,(J)}$.

Why the Romano–Wolf Stepdown Controls FWER: Proof Sketch

- Same template as Holm: identify the *first true null that would be rejected*, then reduce the false-rejection event to a level-α test of the intersection of the true nulls.
- For any subset K of hypotheses, let

$$M_K = \max_{j \in K} T_j,$$

and suppose the intersection null H_K is tested by rejecting when

$$M_K > c_K.$$

Assume the critical values are monotone: if $I \subset K$, then

$$c_I \leq c_K.$$

Why the Romano–Wolf Stepdown Controls FWER (cont'd)

- Fix a data-generating process P, let I denote the set of true nulls under P, and let $\hat{k}$ be the first step at which the procedure rejects a true null.
- At step $\hat{k}$, no true null has yet been rejected, so all true nulls are still in the remaining set $S_{\hat{k}}$:

$$I \subseteq S_{\hat{k}}.$$

- If a true null is rejected at step $\hat{k}$, then the current statistic must exceed the stagewise critical value, which implies

$$\max_{j \in I} T_j > c_I$$

  by monotonicity.
- Therefore,

$$P\big(\text{at least one false rejection}\big) \leq P\left(\max_{j \in I} T_j > c_I\right) \leq \alpha,$$

  because the intersection test for the true nulls is level α.
- So, once the stagewise intersection tests are valid and the critical values are monotone, the Romano–Wolf stepdown procedure controls FWER.

Romano–Wolf: Imposing the Null

- A key step is generating resamples **under the null**. This is the same restricted-resampling principle from the bootstrap lectures: construct the resamples so that the null hypothesis holds by design.
- In an RCT: resample according to the original assignment mechanism (and within any blocks/strata, if applicable). Under a sharp null, this gives the exact randomization distribution.
- In regression settings: use a restricted wild bootstrap based on the null-imposed model.
- In the PROGRESA application, the clean implementation is to reuse the same common permutation draws already used for the individual p-values, but now examine the *joint* distribution of the full vector of grade-specific test statistics.

Romano–Wolf: Properties

Theorem (Romano and Wolf (2005, *JASA*))

If the resampling method consistently estimates the relevant joint distributions used in the stagewise intersection tests, then the Romano–Wolf step-down procedure controls FWER asymptotically.

- Asymptotically controls FWER under general dependence when the resampling approximation is valid.
- Under additional regularity conditions, each false null is rejected with probability tending to one.
- By using resampling to account for the dependence structure, Romano–Wolf can reject more hypotheses than Holm's in finite samples.
- The gain over Holm depends on how much exploitable dependence remains across the studentized test statistics. When that dependence is weak, the gain may be modest.

Romano–Wolf vs. Holm: Comparison

	Holm	Romano–Wolf
Controls FWER	Yes (strong, any dependence)	Yes, asymptotically when resampling is valid
Uses dependence structure	No	Yes (via resampling)
Step-down cutoff	Bonferroni-type	Resampled max over remaining set
Power when dependence is informative	Often lower	Can be higher
Requires resampling	No	Yes
Sensitivity to family definition	Yes	Yes
Implementation (R)	`p.adjust(method = "holm")`	Usually coded directly from common resamples

Both procedures remain sensitive to how the family of hypotheses is defined. Adding irrelevant hypotheses to the family can reduce power for the hypotheses of primary interest.

Romano–Wolf: Choice of Test Statistic

- The algorithm requires test statistics t_j where larger values indicate stronger evidence against the null.
- In principle, one can work with raw statistics (e.g., $\left|\overline{Y}_{1j} - \overline{Y}_{0j}\right|$) or with a monotone transformation such as negative p-values.
- In practice, studentized statistics are often preferable: different hypotheses can have test statistics with different standard errors, so using raw statistics can make the max statistic disproportionately influenced by the noisiest cells.
- Studentization puts the hypotheses on a more comparable scale, so the Romano–Wolf max-t step-down is usually more stable and easier to interpret.
- In the PROGRESA example, this matters empirically: with the unstudentized statistic, Romano–Wolf can sometimes be more conservative than Holm.

Romano–Wolf: Studentized Statistic in PROGRESA

- For hypothesis j, let v index villages, let $D_v \in \{0, 1\}$ denote the village-level treatment indicator, define

$$I_{vj} = \{i : i \text{ is a sampled child in village } v \text{ and cell } j\},$$

and let $n_{vj} = |I_{vj}|$. For villages with $n_{vj} > 0$, define

$$\bar{Y}_{vj} = \frac{1}{n_{vj}} \sum_{i \in I_{vj}} Y_{iv}.$$

- Following Bugni et al. (2025), consider the weighted regression

$$\bar{Y}_{vj} = \alpha_j + \tau_j D_v + u_{vj}, \qquad \text{weighted by } n_{vj},$$

and let $\hat{\tau}_j$ denote the resulting WLS estimate of τ_j, and let $\hat{s}_j$ denote the corresponding sandwich standard error.

- Equivalently,

$$\hat{\tau}_j = \frac{\sum_v D_v n_{vj} \bar{Y}_{vj}}{\sum_v D_v n_{vj}} - \frac{\sum_v (1 - D_v) n_{vj} \bar{Y}_{vj}}{\sum_v (1 - D_v) n_{vj}},$$

so $\hat{\tau}_j$ is the same child-weighted difference in means as before, now written as a weighted difference in village-level means.

Romano–Wolf: Studentized Statistic in PROGRESA (cont'd)

- We use the studentized test statistic

$$T_j = \frac{|\hat{\tau}_j|}{\hat{s}_j}.$$

- Why studentize?
  - Different hypotheses can have test statistics with different standard errors.
  - Studentization makes the max statistic more comparable across hypotheses.
  - Without studentization, the max statistic can be disproportionately influenced by the noisiest cells.
  - In this example, using a studentized statistic gives a more sensible Romano–Wolf correction than using raw differences in means.

Romano–Wolf: Implementation

- Generate one common bank of B village-level treatment reassignments under the sharp null.
- For each draw b and each hypothesis j, recompute the weighted treatment effect $\hat{\tau}_j^{*(b)}$, its standard error $\hat{s}_j^{*(b)}$, and the studentized statistic

$$T_j^{*(b)} = \frac{\left|\hat{\tau}_j^{*(b)}\right|}{\hat{s}_j^{*(b)}}.$$

- For each step-down set S_k, compute the stagewise p-value from

$$\max_{j \in S_k} T_j^{*(b)}, \qquad b = 1, \ldots, B.$$

- Take cumulative maxima of the stagewise p-values to obtain Romano–Wolf adjusted p-values.
- No extra resampling is needed beyond the original permutation exercise; the crucial requirement is that all hypotheses use the *same* permutation draws.

Example: PROGRESA with Romano–Wolf

Table 0.3: Studentized P-Values: Boys, Testing Sharp Null of No Treatment Effect

	No Correction	Bonferroni	Holm's	Romano–Wolf
1	0.38	1.00	1.00	0.83
2	0.02	0.16	0.12	0.16
3	0.69	1.00	1.00	0.83
4	0.00	0.01	0.01	0.01
5	0.32	1.00	1.00	0.83
6	0.00	0.04	0.04	0.04
7	0.01	0.15	0.12	0.13
8	0.56	1.00	1.00	0.83
9	0.26	1.00	1.00	0.83
10	0.26	1.00	1.00	0.83

[*Statistic:* $T_j = \left|\hat{\tau}_j\right| / \hat{s}_j$. *Rows are labeled as grades 1–10; equivalently, these correspond to hgc1 values 0–9.*]

Example: PROGRESA with Romano–Wolf

Table 0.4: Studentized P-Values: Girls, Testing Sharp Null of No Treatment Effect

	No Correction	Bonferroni	Holm's	Romano–Wolf
1	0.63	1.00	1.00	0.99
2	0.92	1.00	1.00	0.99
3	0.35	1.00	1.00	0.92
4	0.04	0.44	0.35	0.31
5	0.10	0.98	0.68	0.55
6	0.01	0.15	0.14	0.13
7	0.00	0.00	0.00	0.00
8	0.61	1.00	1.00	0.99
9	0.69	1.00	1.00	0.99
10	0.68	1.00	1.00	0.99

[*Statistic:* $T_j = |\hat{\tau}_j| / \hat{s}_j$. *Rows are labeled as grades 1–10; equivalently, these correspond to hgc1 values 0–9.*]
[*Interpretation:* After studentization, Romano–Wolf is close to Holm here, suggesting limited exploitable dependence across the studentized test statistics.]

Multiple Hypothesis Testing, Data Mining, p-hacking

Multiple hypothesis testing is a pervasive problem in many fields, e.g.,

- In finance, search for "alpha," testing for whether some funds/managers "beat the market" when searching over many funds/managers;

- Early childhood intervention, e.g.,
  - when testing for effects of intervention over many possible outcomes;
  - when testing for effects of intervention within many different subgroups.

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Related issues

- p-hacking – choosing alternative, significance level, which tests to report, etc., for desired conclusion . . .
- publication bias.

Related to "Replication Crisis"?

Can control for data-mining if formalized and transparent, not if informal and/or hidden.